

power levels. Armstrong oscillator based converter topology is shown in Fig. 5. For such a system, energy source, which is an antenna, has a large internal impedance (up to several kilo-ohms).

Working of the self-oscillating Armstrong converter topology is described below:

Oscillator. It is formed by junction field effect transistor (JFET) and amplifies gate input signal. Oscillator

feedback loop is formed with two inductors, which are coupled, and gate-source capacitor C_{gs} .

Rectifier. To rectify JFET's gate-oscillating voltage, gate-source PN junction of JFET is used. When JFET's gate voltage becomes greater than diode-threshold voltage, the diode is turned on and circuit supplies output to load. When the converter is connected to an energy source, current rises in the primary winding. Secondary winding provides a positive gate voltage on n-channel JFET.

Here, it is assumed that JFET is already on. So C_{gs} connected to gate terminal of JFET is charged to a positive (V_{gs}) value. Gate source PN junction of JFET is forward-biased because of enough gate supply voltage given by the capacitor; output capacitor (C_{out}) is charged with a negative voltage.

Output voltage is therefore negative. When primary current reaches saturation, primary winding voltage gets cancelled and output capacitor delivers a negative voltage to JFET gate to make it pinch off.

Current in the primary winding decreases and negative voltage is applied to JFET gate,

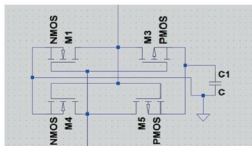


Fig. 4: Bridge rectifier using MOSFET

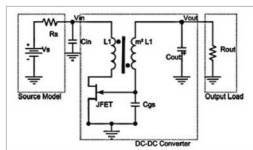


Fig. 5: Self-oscillating Armstrong converter topology

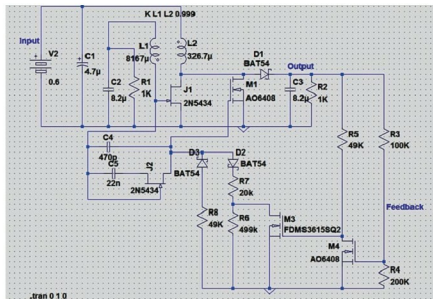


Fig. 6: Self-oscillating ultra-low-voltage step-up converter

which finally switches off the JFET. This peak voltage that switched off JFET falls back to zero, that is, secondary winding current reaches zero value and gate terminal gets rid off the negative voltage earlier applied across it.

Again, JFET gets turned on and the procedure repeats, thus oscillation process gets started.

Main advantage of this circuit is in ultra-low-voltage applications. This circuit has a forward mode of operation. Energy is transferred to output during on phase, which avoids the collapse of the switch-gate voltage oscillations during off-phase (a case of fly-back operation mode).

The self-oscillating ultra-low-voltage converter circuit simulated in LTSpice is shown in Fig. 6. For simulation:

1. Download LTSpice version 4 for Windows or MAC OS from the www.linear.com/designtools/software/#LTSpice
2. After installing the setup file, click on `ultra_stepup_converter.asc` file included in the DVD provided with this month's EFY Plus issue. It will automatically open in the software and you will find the schematic capture window (Fig. 6).
3. Run the schematic by clicking the icon on the left-hand corner and you will get the results.

A self-oscillating converter is capable of boosting 0.6V to approximately 5V. Output power is around 22mW and input power around 23.4mW. Simulation output of the system is shown in Fig. 7. Overall converter efficiency is about 92 per cent in simulation environment.

Energy storage. Energy storage will supply power to the sensor, microcontroller or data transceiver module. Advantages of the super capacitor make it ideal for energy storage. However, it takes a lot of time to charge and discharge.